

LCS6260 AT Command Set

Release Notes

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Instruction set syntax

AT commands use command lines based on ASCII code. The command ends with carriage return and line feed. The command format is as follows: The request message format is: AT+<CMD>[OPTION][para, ...] < > Required content []

Optional options

domain	
AT+	Description Command
CMD	message prefix
Command string OPTION	Command operator. It can be the following: "=": indicates setting parameters. "?": indicates querying the current value of the parameter. "": indicates executing the command. "=?": indicates requesting the Help of the command
for \r\n	Usage information indicates
	the parameter value to be set Carriage return and line feed characters,

ASCII 0x0d, 0x0A respectively Table 1 AT request message format

The response message format is: OK\r\n (command executed successfully)

ERROR\r\n (Command execution failed, or the input AT command is wrong)

For

example: 1. Execute the module restart command:

AT+RST 2. Set the module's Wi-Fi working mode to Station: AT+CWMODE=1 2. Query the

module's Wi-Fi current working mode: AT+CWMODE? 3. View the module's Wi-Fi

working mode help information: AT+CWMODE=?



Note: The

LCS6260 module uses UART0 for AT command operation. • The default baud

rate is 115200. • Not all AT commands

support the format described in Table 1. Please refer to the specific command description. • The angle brackets

<> and square brackets [] in the message format are for illustration only and are not included in the actual operation instructions. •

String parameters need to be included in double quotes, for example: AT+CWSAP="LCS6260","12345678",6. • AT commands must be

capitalized and end with a new line (CR-LF).

AT commands

AT -- AT command verification

execution instruction AT	
Response Description	
OK Usage Example AT	

ATE -- switch command echo function

execution command ATE	
Response Description	
OK Parameter Description • ATE0: Disable echo • ATE1: Enable	
echo Example ATE1 // Enable echo Notes	
Echo is enabled by default	

AT+RST -- Software reset

Execution command AT+RST	
Response description	
OK Usage example AT+RST	

AT+VER --Query software version information

query command AT+VER?	
Response description +VER:V003 //Return software version information OK	
Example of using AT+VER?	

AT+CWRADIO -- Wireless RF switch status command

description query command: AT+CWRADIO? Function: Query the current RF switch status	Setting Instructions AT+CWRADIO=<en> Function: Set the current RF switch status
Response Description +CWRADIO:<en> OK	OK or ERROR
Parameter Description • <en>: Indicates whether the Wi-Fi radio is on or off ÿ 0: en is 0, which means it is off ÿ 1: en is 0, which means it is on Example AT+CWRADIO?	

AT+CWMODE --Set the current working mode of Wi-Fi

Command Description View help information Command: Query command: AT+CWMODE=? AT+CWMODE?	Set the command: AT+CWMODE=<mode>
--	--------------------------------------

Response description	OK	+CWMODE:<mode> //Value list OK	OK
Parameter Description	• <mode>: ý 1: Station mode ý 2: SoftAP mode ý 3: SoftAP+Station mode Usage		
example AT+CWMODE=3			

AT+CWJAP --Connect to AP hotspot Command

Description Query command:	AT+CWJAP? Function: Query the information of the AP that the Station is connected to	Set the command: AT+CWJAP=<ssid>,<pwd> Function: Set the AP that the Station needs to connect to
Response description	OK	WIFI CONNECTED //Station is connected to AP WIFI GOT IP //Station obtains the IP assigned by AP OK or ERROR
Parameter Description	• <ssid>: SSID of the target AP • <bssid>: MAC address of the target AP • <channel>: channel • <rssi>: signal strength indicator	• <ssid>: string parameter, SSID of the target AP • <pwd>: string parameter, the password is up to 64 bytes in ASCII format
value Example	AT+CWJAP="abc_ap","12345678" Note: Parameter setting	
requires	Station mode to be enabled. If ssid or pwd contains special symbols, such as escaping, other characters are invalid. . or " or \,	

AT+CWSAP --Configure SoftAP parameters Command

Description Query command:	AT+CWSAP? Function: Query SoftAP configuration parameters	Set the command: AT+CWSAP=<ssid>,<pwd>,<chl> Function: Set SoftAP configuration parameters
Response Description	OK	OK or ERROR
Parameter Description	• <ssid>: string parameter, the name of the hotspot • <pwd>: string parameter, password length range: 8-64 bytes • <chl>: channel number	
Example AT+CWSAP="LCS6260","1234567890",6	Note: Parameter setting requires	
SoftAP mode to	be enabled. If ssid or pwd contains special symbols, such as escape, other characters are invalid. . or " or \,	

AT+CWLAP -- Scan currently available APs

Execute the command	AT+CWLAP
Response Description	OK
Parameter Description	• <auth>: authentication mode.

	ý 0: OPEN ý 1: WEP ý 2: WPA_PSK ý 3: WPA2_PSK ý 4: WPA_WPA2_PSK • <ssid>: string parameter, AP's SSID • <rssi>: signal strength • <bssid>: string parameter, AP's MAC address • <channel>: channel number Example AT+CWLAP

AT+CWQAP --Disconnect from **AP** . Execute

command AT+CWQAP	
Response Description	WIFI DISCONNECT //Wi-Fi disconnects from AP OK
Example of using	AT+CWQAP

AT+CWSMARTCONFIG -- Wi-Fi smart network configuration

command AT+CWSMARTCONFIG=<en>	Response
description	OK or ERRORParameter
description • <en>: ý 0:	Stop smart network configuration ý 1: Enable
smart network configuration	Example AT+CWSMARTCONFIG=1

AT+CIUPDATE --Firmware remote upgrade

execution command AT+CIUPDATE	
Response Description	
OK Usage	Example AT+CIUPDATE

AT+CIPSTAMAC --Query the **MAC** address of the **LCS6260 Station** interface .

AT+CIPSTAMAC?	
Response Description	+CIPSTAMAC:<mac> OK
Parameter Description	<mac>: string parameter, MAC address of LCS6260 Station
Usage Example	AT+CIPSTAMAC?

AT+CIPSTATUS --Display network connection status information

Execute the command AT+CIPSTATUS	
Response Description	+CIPSTATUS: <link ID>, <type>, <remote IP>, <remote port>, <local port>, <tetype> Parameter
Description • <link ID>	Network connection ID (0~4), used for multiple connections

	• <type>: string parameter, "TCP" or "UDP" • <remote IP>: string parameter, remote IP address or domain name • <remote port>: remote port number • <local port>: LCS6260 local port value • <tetype>: ý 0: LCS6260 as client ý 1: LCS6260 as server Example of use AT+CIPSTATUS

AT+CIPDOMAIN --Domain name resolution function

setting command	AT+CIPDOMAIN=<domain name>
Response Description	+CIPDOMAIN: <IP address>
Parameter Description	• <domain name>: The domain name to be resolved
resolved Example	Example of use AT+CIPDOMAIN="www.baidu.com"

AT+CIPMUX --Network multi-connection configuration

command description	query command: AT+CIPMUX?	Set the command: AT+CIPMUX=<mode>
Response Description	+CIPMUX: <mode> OK	OK
Parameter Description	• <mode>: ý 0: Single connection ý 1:	
Multiple connections Usage Example	AT+CIPMUX=1 Note 1. The default is single connection 2. Only in non-transparent mode (AT+CIPMODE=0) can you set multiple connections. 3. You must set the connection mode when there is no connection established. 4. If a TCP server is established and you want to switch to a single connection, you must shut down the server (AT+CIPSERVER=0). Only supports multiple connections	

AT+CIPMODE --Configure network transmission mode

Command Description	Query command: AT+CIPMODE?	Set the command: AT+CIPMODE=<mode>
Response Description	+CIPMODE:<mode> OK	OK
Parameter Description	• <mode>: ý 0: non-transparent mode ý 1: transparent mode, only supports TCP single connection and UDP fixed communication peer	
Example	AT+CIPMODE=1 Note: Enter +++ to exit	
transparent transmission mode		

AT+CIPSTART -- Establish TCP client connection or UDP transmission to

establish TCP

connection Command	Description TCP single connection (AT+CIPMUX=0): AT+CIPSTART=<type>,<rmIP>,<rmport> Response Description	TCP multi-connection (AT+CIPMUX=1): AT+CIPSTART=<linkID>,<type>,<rmIP>,<rmport>
OK or ERROR	If the connection already exists, ALREADY CONNECT is returned	
Parameter Description	<link ID>: Network connection ID (0~4), used for multiple connections • <type>: string parameter, the connection type is "TCP" or "UDP" • <rmIP>: string parameter, remote IP address or domain name • <rmport>: remote port number Example	
AT+CIPSTART="TCP","xx.cn",8000 Establish UDP transmission Command		

Description UDP single

connection (AT+CIPMUX=0): UDP multiple connections (AT+CIPMUX=1): AT+CIPSTART=<linkID>,<type>,<rmIP>,<rmport>,<lcport>	AT+CIPSTART=<type>,<rmIP>,<rmport>,<lcport>	
Response Description	OK or ERROR If the connection already exists, ALREADY CONNECT is returned	
Parameter Description	• <link ID>: Network connection ID (0~4), used for multiple connections • <type>: string parameter, the connection type is "TCP" or "UDP" • <rmIP>: string parameter, remote IP address or domain name • <rmport>: remote port number • <lcport>: for UDP transmission, set the local port number Example	
AT+CIPSTART="UDP","192.168.4.2",8000,1002 Note To establish UDP transmission, you must fill in the local port number		

AT+CIPCLOSE --Close TCP /UDP connection

Command Description	Setting command (for multi-connection mode): AT+CIPCLOSE =<link ID>	Execute command (for single connection mode): AT+CIPCLOSE
Response Description		
OK Parameter Description	• <link ID>: The connection ID to be closed. When ID=5, all connections are closed (the ID of 5 is invalid after the server is enabled)	
	Example AT+CIPCLOSE	

AT+CIPSERVER -- TCP server configuration

command AT+CIPSERVER=<mode>[,<port>]	Response description OK
Parameter description	
• <mode>:	• 0: Close TCP server • 1: Establish TCP server • <port>: port number. If no parameter is set, the default port number is 333.
Example of using	AT+CIPMUX=1 AT+CIPSERVER=1 Note 1: TCP
server can be established only in network multi-connection mode (AT+CIPMUX=1)	

	2. After establishing the TCP server, automatically establish the TCP server listening 3. When a TCP client accesses, it will automatically occupy a connection ID
--	--

AT+CIPSTO --Set TCP server timeout

Command Description Query	Set the command:
command: AT+CIPSTO?	AT+CIPSTO=<time>
Response description +CIPSTO:<time> OK	OK
Parameter Description • <time>: TCP server timeout, value range 0 ~ 7200, unit: seconds	
Usage Example AT+CIPMUX=1 AT+CIPSERVER=1,1001 AT+CIPSTO=10 Note: 1. As	
a TCP server, it will disconnect the TCP client connection that has not communicated until it times out. 2. If you set AT+CIPSTO=0, it will never time out. This setting is not recommended.	

AT+CIPSEND --Network data sending

command description Setting	Execute the command: AT+CIPSEND Function: Start sending data in transparent transmission mode
command: 1. Single connection (+CIPMUX=0) AT+CIPSEND=<len> 2. Multiple connections (+CIPMUX=1) AT+CIPSEND=<link ID>,<len> 3. When using UDP transmission, you can set the remote IP and port AT+CIPSEND=[<linkID>,<len>[,<rmIP>,<rmport>] Function: In non-transmission mode, set the length of the sent data	
Response Description Send data of the specified length. After receiving this command, it will first return to >, then start receiving serial port data. When the data length reaches <length>, it will send data, return to normal command mode, and wait for the next AT command. If the connection is not established or the connection is disconnected, the response is returned: ERROR If the data is sent successfully, the response is returned: SEND OK	After receiving this command, return to the line >. Enter transparent transmission mode to send data, maximum 2048 per packet Bytes, or each packet of data at intervals of 50 ms or more When a single packet of +++ is input, it returns to the normal AT command model. When sending +++ to exit transparent transmission, please wait at least 1 second before sending again Next AT command. This command must be used in transparent transmission mode and single connection.
Parameter Description • <link ID>: Network connection ID (0~4), used for multiple connections • <len>: Numeric parameter, indicating the length of the data to be sent, the maximum length is 2048 bytes. • [<rmIP>]: You can set the peer IP or domain name for UDP transmission transmission • [<rmport>]: You can set the peer port number for UDP transmission	

AT+CIPSNTPCFG --Set the sntp time domain and sntp server

Command Description Query	Set the command:
command: AT+CIPSNTPCFG?	AT+CIPSNTPCFG=<tz>,<period>,<server>

Response description +CIPSNTPCFG: <tz>,<period>,<server> OK	OK
Parameter Description • <tz>: time domain, range [-13,13]. • <period>: sntp period, in seconds •	
<server>: sntp server. Example AT+CIPSNTPCFG=8,3600,"pool.ntp.org"	
Note: The default time domain setting is 8, the default period is 3600, and the default sntp server is "pool.ntp.org".	

AT+CIPSNTPTIME --Query the sntp time

query command AT+CIPSNTPTIME?	
Response description +CIPSNTPTIME:"Thur 2019-06-06 15:14:07" OK	
Example of using AT+CIPSNTPTIME?	

Other

AT command usage examples

This section introduces some examples of how to use LCS6260 AT commands.

Basic AT command examples

1. LCS6260 software resets the serial port

and sends execution instructions:

AT+RST

serial port response:

OK 2.

Query the software version information. Send

query command through the serial port:

AT+VER?

Serial port response:

+VER: V003

OK

Wi-Fi AT Command Examples

1. Configure Wi-Fi working mode.

Send setting command through

serial port: **AT+CWMODE=1 //Set Wi-Fi working mode to Station.** Serial port

response: OK.

Send

query command through serial

port: **AT+CWMODE?**

Serial port response:

+CWMODE:1

OK 2.

Scan and send execution

commands through the AP serial port:

AT+CWLAP serial

port response:

+CWLAP:3,"LCS6260_WiFi02",-58,"30:ae:a4:03:c5:ad",9

+CWLAP:4,"JOE",-68,"ec:26:ca:d0:58:b4",6

.....

OK

3. Configure the LCS6260 module to initiate the SoftAP serial

port to send the setting command:

AT+CWMODE=2

AT+CWSAP="LCS6260_01","1234567890",6Serial port response: OKSerial

port sends

query

command: **AT+CWSAP?**

Serial port response:

```
+CWSAP:"LCS6260_01","1234567890"
OK
```

4. Connect to AP hotspot

serial port and send setting command:

```
AT+CWJAP="LCS6260_ap","1234567890" Serial port response:
```

```
OK 5.
```

Disconnect the AP and send the

execution command through the serial port:

```
AT+CWQAP serial
```

port response:

```
OK
```

Other

TCP/IP network communication related AT command examples

TCP network communication

1. TCP client single connection operation • Set

Wi-Fi working mode Serial port sends

setting instructions:

```
AT+CWMODE=1 Serial port //Station mode
```

response:

```
OK •
```

Connect to AP hotspot

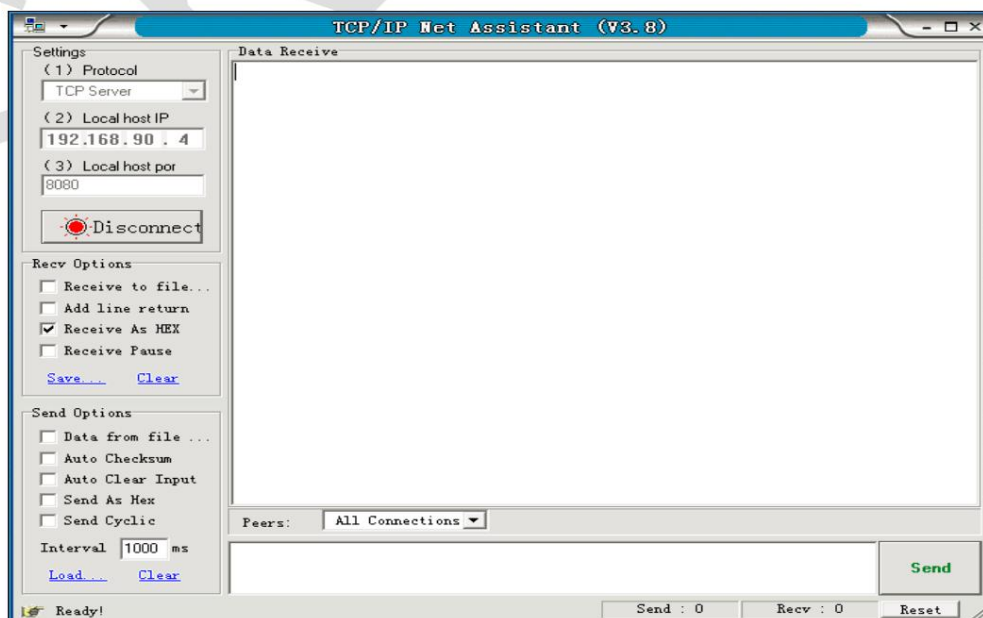
serial port and send setting command:

```
AT+CWJAP="ssid","password" //Router hotspot name and password Serial port response:
```

```
OK
```

• Connect the PC and LCS6260 module to the same router hotspot, and then use the network communication debugging assistant tool on the PC to establish a

TCP server



- After connecting to the router hotspot, the LCS6260 module establishes a TCP client to connect to the server

Serial port sends setting instructions:

AT+CIPSTART="TCP","192.168.90.4",8080 //protocol, server IP & port serial port response:

OK •

Network sends data

Set the network transmission mode to non-transparent mode, and then specify the length of the data to be sent:

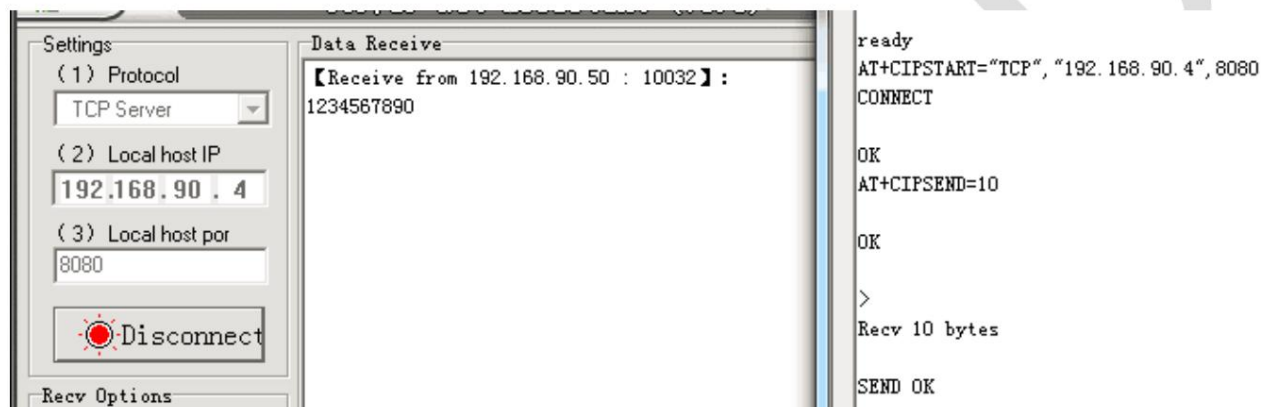
AT+CIPMODE=0 //Set to non-transparent mode

AT+CIPSEND=10 > //Set the length of the data to be sent, here is 10 bytes

1234567890 Serial //Enter the data to be sent

port response:

SEND OK



Set the network transmission mode to transparent mode, and then execute the send command to send data:

AT+CIPMODE=1

AT+CIPSEND serial

port response:

> // Respond to the prompt and enter the data to be sent

When you need to exit the transparent transmission mode, the serial port sends a string of +++ to exit the transparent transmission mode and enter the normal command mode.



Notice:

- The purpose of ending the data packet with +++ is to exit the transparent transmission mode and enter the normal AT command mode.
Stay connected.

- Close the connection between the TCP client and the TCP server

The serial port sends execution instructions:

AT+CIPCLOSE serial

port response:

CLOSED

OK

2. TCP server establishes multiple connections

- Set network communication to multi-connection mode

Serial port sends setting instructions:

AT+CIPMUX=1 Serial

port response:

OK

- Establishing a TCP server

Serial port sends setting instructions:

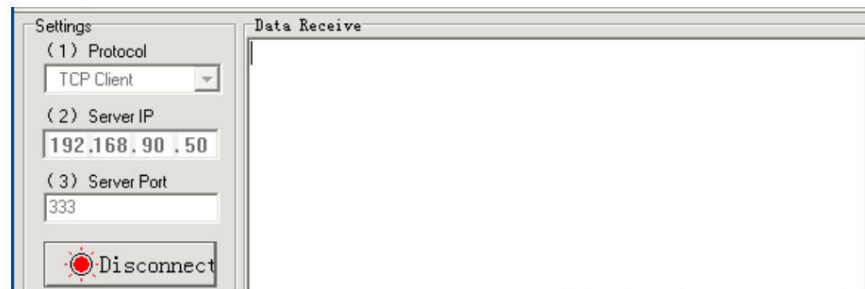
AT+CIPSERVER=1 // If the port number is not filled in, the default port 333 is used. Serial port response:

OK

- The PC and LCS6260 module are connected to the same router hotspot. After the LCS6260 is connected to the router hotspot, it obtains the router hotspot information.

The IP address assigned by the router is 192.168.90.50. The PC uses the network communication debugging assistant tool to establish a TCP client.

And set the server IP to 192.168.90.50 and the server port number to 333, as shown in the following figure:



- Send network data operation:

AT+CIPSEND=0,10 >
1234567890 Serial

//The connection ID of the first connection is 0

//Specify the client with connection ID 0 to send data with a length of 10 bytes

//Enter the data to be sent

port response:

SEND OK •

LCS6260 receives network data:

+IPD,0,n:xxxxxxxx //Receive n bytes of data from the client with connection ID 0. The data content is xxxxxxxxxx

- Specify the client connection ID to disconnect the TCP connection with the client

Serial port sends setting instructions:

AT+CIPCLOSE=0 Serial

//Disconnect the TCP connection with the client whose connection ID is 0

port response:

0,CLOSED

OK

UDP transport

Create UDP transmission through AT+CIPSTART command

- Set Wi-Fi working mode

Serial port sends setting instructions:

AT+CWMODE=1 Serial

//Station mode

port response:

OK

- Connect to an AP hotspot

Serial port sends setting instructions:

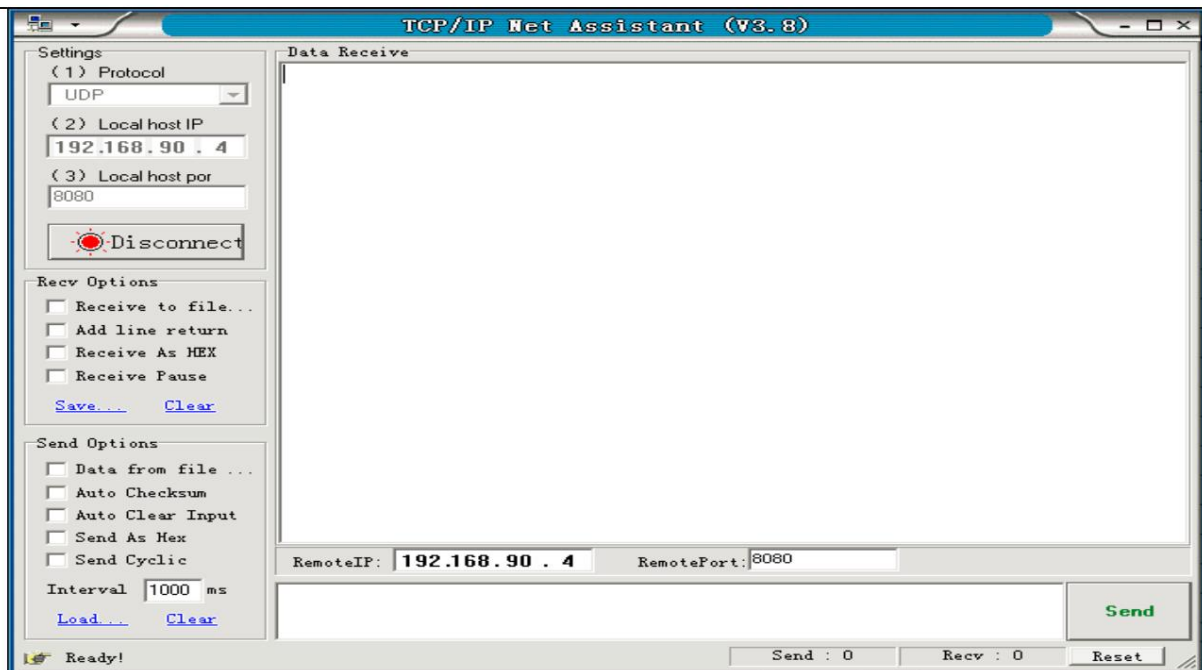
AT+CWJAP="ssid","password" //AP hotspot name and password

Serial port response:

OK

- Connect the PC and LCS6260 module to the same router hotspot, and then use the Network Communication Debug Assistant tool on the PC.

To establish a UDP communication



- Establishing multiple connection operations

Serial port sends setting instructions:

AT+CIPMUX=1 Serial

port response:

OK

- Establish UDP transmission and set the connection ID to 4

Serial port sends setting instructions:

AT+CIPSTART=4,"UDP","192.168.90.4",8080,1112

Serial port response:

4,CONNECT

OK



Notice:

- "192.168.90.4" and 8080 are the remote IP and port for UDP transmission, which is the UDP configuration set by the PC.
- Local port number 1112 is the local port for LCS6260 module to establish UDP transmission. Users can customize this port. Slogan. If not defined beforehand, the value of this parameter will be random.

- Send network data

operation:

AT+CIPSEND=4,10

//Specify the connection ID as 4 to send data with a length of 10 bytes

//Enter the data to be sent

> 1234567890 Serial port response:

SEND OK •

LCS6260 receives network data:

+IPD,4,5:hello //Receive 5 bytes of data from the client with connection ID 4, the data content is hello

- Close the network connection with the specified connection ID

Serial port sends setting instructions:

AT+CIPCLOSE=4 Serial

port response:

4,CLOSED

OK

AT command prompt information

Tips	describe
AT Ready	Indicates that the AT command set is ready and can be used to send commands
ERROR	AT command execution response error or no AT command
WIFI CONNECTED	Station is connected to SoftAP
WIFI GOT IP	Station has obtained the IP address assigned by SoftAP
WIFI DISCONNECT	Station disconnects from Soft AP
<linkID>,CONNECT	The network communication link of the connection ID number has been successfully connected
<linkID>, CLOSED	The network communication link of the connection ID number has been successfully disconnected
+IPD	Network data receiving prompt
Other	...

contact information



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